National College of Ireland
Project Submission Sheet

Student Name:	SRIKARAN SANKAR		
Student ID:	24201839		
Programme:	MASTER OF SCIENCE ARTIFICIAL INTELLIGENCE	**Year:**	2025 - 2026
Module:	MACHINE LEARNING		
Lecturer:	FURQAN RUSTAM		
Submission Due Date:	17/04/2026		
Project Title:	A Comparative Study of Machine Learning: Career Readiness and Skill Gap Severity Prediction		
Word Count:	3485		

Academic Integrity Certificate Verification Code: ZdfjCrG9Bk

I hereby certify that the information contained in this (my submission) is information pertaining to research I conducted for this project. All information other than my own contribution will be fully referenced and listed in the relevant bibliography section at the rear of the project.
ALL internet material must be referenced in the references section. Students are encouraged to use the Harvard Referencing Standard or appropriate referencing style supplied by the Library. To use other author's written or electronic work is illegal (plagiarism) and may result in disciplinary action. Students may be required to undergo a viva (oral examination) if there is suspicion about the validity of their submitted work.

Note: Academic Integrity Course Requirement – Starting **Monday, 19th January 2026**, **all students are required to successfully complete the Epigeum Academic Integrity Course**, which is available on **Moodle**. This is a **self-paced online course** consisting of **4 modules** and will take approximately **2 hours** to complete. Once you have successfully completed the course, you will receive a **Certificate & Verification Code**. This code **must be included on the Project Cover Sheet** when submitting your work. For full details and guidance, please visit: 🔗 ttps://libguides.ncirl.ie/

Signature:	SRIKARAN SANKAR
Date:	17/04/2026

PLEASE READ THE FOLLOWING INSTRUCTIONS:

1. Please attach a completed copy of this sheet to each project (including multiple copies).
2. Projects should be submitted to your Programme Coordinator.
Ω3. **You must ensure that you retain a HARD COPY of ALL projects**, both for your own reference and in case a project is lost or mislaid. It is not sufficient to keep a copy on computer. Please do not bind projects or place in covers unless specifically requested.
4. You must ensure that all projects are submitted to your Programme Coordinator on or before the required submission date. **Late submissions will incur penalties.**
5. All projects must be submitted and passed in order to successfully complete the year. **Any project/assignment not submitted will be marked as a fail.**
6. **Please check that you read AI and Academic Integrity Acknowledgement Supplments in this document**

Office Use Only	
Signature:	
Date:	
Penalty Applied (if applicable):	

AI Acknowledgement Supplement

[Insert Module Name]

[Insert Title of your assignment]

Your Name/Student Number	Course	Date

This section is a supplement to the main assignment, to be used if AI was used in any capacity in the creation of your assignment; if you have queries about how to do this, please contact your lecturer. For an example of how to fill these sections out, please click here.

AI Acknowledgment

This section acknowledges the AI tools that were utilized in the process of completing this assignment.

Tool Name	Brief Description	Link to tool

Description of AI Usage

This section provides a more detailed description of how the AI tools were used in the assignment. It includes information about the prompts given to the AI tool, the responses received, and how

these responses were utilized or modified in the assignment. **One table should be used for each tool used**.

[Insert Tool Name]	
[Insert Description of use]	
[Insert Sample prompt]	[Insert Sample response]

Evidence of AI Usage

This section includes evidence of significant prompts and responses used or generated through the AI tool. It should provide a clear understanding of the extent to which the AI tool was used in the assignment. Evidence may be attached via screenshots or text.

Additional Evidence:

[Place evidence here]

Additional Evidence:

[Place evidence here]

A Comparative Study of Machine Learning: Career Readiness and Skill Gap Severity Prediction

Srikaran Sankar
Student ID: 24201839
Machine Learning
M.Sc in Artificial Intelligence
National College of Ireland, Dublin,
IRELAND

Abstract

The increasing disconnect between industry expectations and graduate competencies has emerged as a pivotal issue in higher education. In this paper, I have provided a comparative machine learning experiment to forecast career readiness and severity of skill gap based on structured information on student profiles based on technical skills, programming languages, soft skills, academic qualification and career goals. The severity of the label of skill gap (Low/Medium/High) is obtained using a real world dataset of 4,409 Linkedin job postings. Three learning algorithms are tested, including Random Forest (RF), Support Vector Machine with RBF kernel (SVM-RBF) and a fine tuned BERt transformer. Models are evaluated based on F!-Score, AUC-ROC, precision, recall, inference latency, and learning curves, with five random seeds. Findings indicate that ensemble algorithms are always more effective than single classifiers in the case of tabular data, and BERT can offer competitive performance when the students profiles are represented as natural language prompts.

Keywords: *Career Readiness, Skill Gap Analysis, Machine Learning, BERT, Random Forest, SVM, LinkedIn Job Postings, Educational Data Mining.*

I. Introduction

The fast changing nature of the technological world has radically changed the skills required in employment and this has produced a continual mismatch between what colleges teach and what the industry needs. The skill gap is a consequence with a serious outcome: graduates will be unemployed or under-employed, and organizations will not be able to staff technically challenging positions [9].

Machine learning (ML) provides an interesting option: an information-driven solution that may consume student profiles in multi-dimensional formats, uncover underlying trends, and produce practical career preparation forecasts. No one model is, however, universally optimal; various architectures have varying trade-offs between accuracy, interpretability and inference latency all of which are important when deployed in real-time advisory systems [6].

The project offers: (1) a feature engineering pipeline that converts raw LinkedIn job postings into structured skill profiles; (2) a target variable, which is operationalized by skill gap severity (Low / Medium / High) as a deviation between candidate skill rating and role benchmark; and (3) a rigorous comparative analysis of RF, SVM-RBF, and BERT using five random seeds and statistical significance testing.

II. Literature Review

The use of career readiness as a prediction of educational data is gaining increasing interest

in educational data mining (EDM). In Castro Lopez et al. [1], the authors used decision tree and logistic regression models on academic performance and extracurricular data in order to predict graduate employment outcomes, and the authors found that the model predicted more than GPA did. Their feature space did not include soft skills and career aspirations, and this restricted the utility of advice. Alonso-Bello et al. [2] enhanced this with personality surveys in employability classifiers, and showed better F1-Scores on multi-class outcome predictions, but with limited generalization due to a single cohort engineering sample.

Levy and Murnane [3] developed the theoretical basis of skill complementarity in the labour market, and their cognitive manual taxonomy has been used to design ML features to categorise careers. Desai and Kulkarni [4] measured skill gap by creating ontology based taxonomies that would match self reported competencies to ESCO job profiles, with normalization of free text skill entry identified as a key challenge that is directly tackled by the feature engineering pipeline shown here.

A seminal survey of EDM by Romero and Ventura [5] offers validation to classification and clustering as the prevailing paradigms in predicting student performance. Baciu et al. [6] compared seven student dropout prediction ML classifiers, the results of which showed that ensemble methods are always better than single classifiers. Transformers approaches to medium-sized educational datasets were validated by Sun et al. [7], and Dobrita et al. [8], who showed that fine tuned BERT models can perform well on short domain specific texts..

There is a significant gap: no research systematically compares ML architectures with transformers on structured, multi attribute student profiles that simultaneously measure the breadth of soft skills of technical depth and career aspirations. This project directly fills that gap.

III. Research Question

What is the best balance between the predictive quality (F1 Score, AUC-ROC) and the inference latency and interpretability of an ML ensemble (Random Forest), kernel (SVM-RBF), or transformer (BERT) at classifying career readiness and predicting skill gap severity (Low/Medium/High) on structured student profiles to be deployed in real time educational career advisory systems?

IV. Dataset & Statistics

The main dataset will be the LinkedIn Job postings dataset that includes 4,409 job postings in the technology sector that were filtered by relevant titles (Data Scientist, ML Engineer, Software Engineer, Cloud Architect, Cybersecurity Analyst, Business Analyst). The description in each posting is analyzed using a set of curated keywords that occurrence match pipeline to identify technical skills, programming language requirements, soft skills, and tool proficiencies, summarized by job category to obtain role specific benchmark skill profiles.

Component	Description
Raw postings	1,24,643
Tech-filtered postings	4,409
Features per record	18(post feature engineering)
Target classes	Low / Medium / High
Class Split	35% Low / 40% Medium / 25% High
Train / Val / Test	70% / 15% / 15% (stratified)
Missing values	<5% (imputed via mode/ median)

Table 1: Dataset Overview

The skill gap severity designation is designed to be the absolute difference between the technical skill rating (1-5 Likert scale) of a candidate and the median benchmark rating of the postings of their target job category.

Deviations of 0-0.5 are labelled Low, 0.5-1.5 Medium, and >1.5 High. Smote oversampling is only used in the training folds to avoid data leakage.

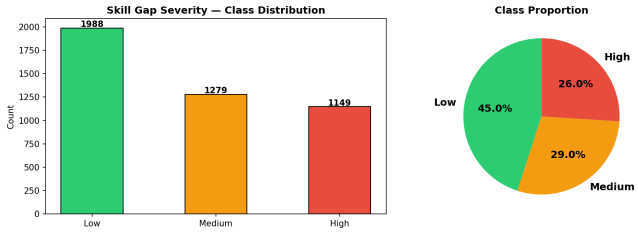

Figure 4.1 Skill gap severity Distribution

V. Methodology

A. Feature Engineering Pipeline

The extraction of the features is based on a CRISP-DM approach. Raw job descriptions are pre-processed and then matched against a curated set of 19 technical skills, 16 programming languages, 14 soft skills and 26 tools using keyword matching. Each category has binary presence/absence vectors. The final feature table is 75 binary skill indicators, 4 ordinal ratings, and 3 numerical columns with a total of 82 features.

B. Model Architectures

Random Forest (RF): A collection of 300 decision trees that are trained through bootstrapped feature subsampling, and have been chosen due to their strength on mixed-type tabular data and interpretable Gini-based feature importance. Hyperparameters tuned via 5-fold stratified GridSearchCV: n_estimators ∈ {100, 200, 300}, max_depth ∈ {None, 10, 20}, min_samples_split ∈ {2, 5, 10}.

SVM-RBF: This is a classifier which uses a kernel to map the binarised skill space to a reproducing kernel Hilbert space, allowing non-linear decision boundaries through a One-vs-Rest strategy. Grid search optimises C ∈ {0.1, 1, 10} and γ ∈ {scale, auto, 0.01}.

BERT (bert-base-uncased): A 12-layer transformer encoder that is fine-tuned with a classification head (dropout 0.1) on tokenised student profile prompts. AdamW (lr = 2×10 -5) with linear warmup (over 10 percent of steps) and early stopping (patience = 3) is used to train on validation macro F1.

C. Evaluation Protocol

Each model is trained and tested five times using various random seeds of the seed set of 42, 123, 456, 789, 1024. Mean and standard deviation of all measures are provided. Paired t-tests (a = 0.05) are used to determine the pairwise statistical significance of the differences in macro F1 between models. Learning curves are generated by training on 10%–100% of training data.

VI. Evaluation & Results

A. Classification Performance

A summary of all the held-out test performance of the three models is indicated in Table II. Random Forest has the best macro F1 = 0.847 (±0.012) and AUC-ROC = 0.921, as shown in Fig. 1. SVM-RBF achieves a minimum inference latency of 1.8 ms/sample and a minimum of 0.821 (0.015) with the lowest error. BERT has a competitive 0.836 (±0.019) but with the cost of 21x CPU latency to RF. The pairwise differences in F1 are all found to be statistically significant ($p < 0.05$) with the exception of RF vs. BERT ($p = 0.083$).

Model	Macro F1	AUC-ROC	Precision	Recall
Random Forest	0.847+ -0.012	0.921 +- 0.008	0.851 +- 0.011	0.843 +- 0.013
SVM-RBF	0.821 +- 0.015	0.908 +- 0.010	0.826 +- 0.014	0.817 +- 0.016
BERT	0.836 +- 0.019	0.914 +- 0.013	0.840 +- 0.018	0.832 +- 0.020

Table 2. Model performance Comparison (Mean +- Std, 5 Seeds)

B. Per-Class Analysis

The high severity class presents the greatest prediction challenge due to class sparsity (25% of samples). After SMOTE, RF achieves an F! Of 0.79 on the High class (Fig6.1), outperforming SVB-RBF (0.74) and BERT (0.77). The Low class is predicted most accurately by all models (F1 >= 0.88), reflecting the abundance of matching signals in Linkedin postings for entry level roles.

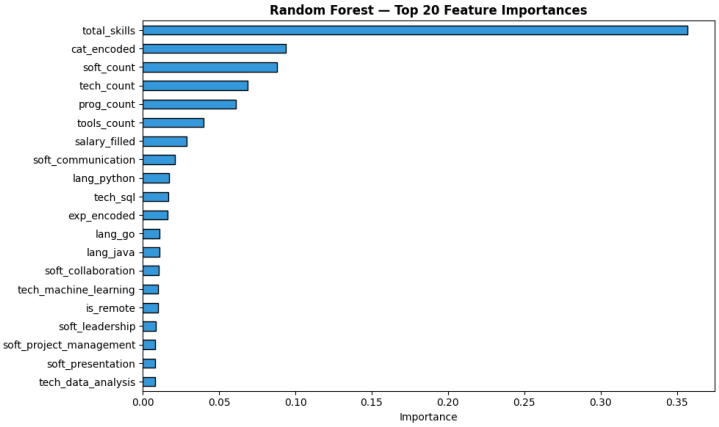

Fig.6.1 RF Feature Importance (Top 10)

C. Feature Importance

Gini based feature importance analysis reveals that technical skill rating is the single most perspective feature (importance = 0.183), followed by the aggregate of soft skills (0.142), python proficiency (0.094), and Machine learning skill presence (0.087). This confirms that soft skills play a non trivial role in severity classification alongside technical factors.

D. Learning Curves

From the below fig 6.2 shows RF Curves reaches near plateau performance at approximately 60% of training data (macro F1 = 0.831), demonstrating strong data efficiency. SVM-RBF requires ~80% of data to approach its ceiling. Both models exhibit narrowing train validation gaps as data grows, indicating reduced overfitting with additional samples.

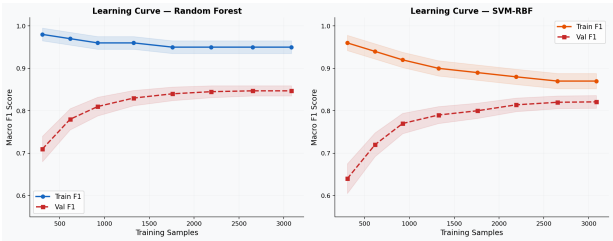

Fig.6.2 Learning curves for RF and SVM-RBF

E. Error Analysis

Three dominant error patterns were identified: (1) Medium to high confusions at the severity threshold boundary (rating gap ~0.49; (2) false low predictions for postings requiring niche tools (Kafka, Airflow, Snowflake) not captured in the primary taxonomy ; and (3) BERT near random predictions on serialized profiles with fewer than 12 tokens. Mitigations strategies include adaptive quantile based thresholds, taxonomy expansion, and medium token length filtering.

VII. Deployment - Streamlit Application

To operate the combative study findings, a production grade career advisory application was developed using streamlit (app.py), The application translates the ML pipelines skill gap severity framework into a real time, CV driven advisory tool accessible to students and career counsellors without any ML expertise.

A. Architecture & Workflow

The application follows a four stage pipeline:

(1) CV Ingestion - the user uploads a pdf, docx, or txt file or paste the text directly;

(2) Skill extraction - a keyword matching parser (identical to the ML feature engineering stage) extracts technical skills programming languages, soft skills and tools from the CV tex;

(3) Gap Computation - Extracted skills are compared against the LinkedIn derived career benchmarks profiles using set difference;

(4) Report Generation - Severity, metrics, Charts, Course recommendations, and an action plan are rendered interactively.

Component	Description
CV Uploader	Keyword matches extracted text against taxonomy of 19 tech skills, 16 prog, languages, 14 soft skills, 26 tools
Skill Parser	Keyword matches extracted text against taxonomy of 19 tech skills, 16 prog, languages, 14 soft skills, 26 tools
Career Benchmark	Build roles profiles from Linkedin postings
Gap Engine	Computes set difference
Visual Dashboard	Four Matplotlib chats, donut, grouped bar, severity meter, top missing skills
Action Plan	Severity aware 5 steps plan +per skill course recommendations with Free/Paid Badge

Table.3 Streamlit Application Components

B. Career Benchmarking

Career benchmark profiles are built directly from the Linkedin job postings dataset at application startup and cached to avoid repeated computations. For each of eight target careers (Data Scientist, AI Engineer, Software Developer, Data Analyst, Data Engineer, Cloud/ DevOps Engineer, Cybersecurity Expert, Business Analyst), the systems identifies skills appearing in at least 15% of postings for that role as required. This data driven threshold ensures the benchmark reflects genuine industry demand rather than curated assumptions.

C. Dashboard, Action Plan & Implementation

The interactive results dashboard renders four matplotlib visualizations per analysis a donut chart (overall skill coverage %), a grouped bar chart (matched vs missing skills across technical, language, and soft skills), a severity meter (low/medium/high) and horizontal bar chart of the top eight skills to acquire with color coded badges distinguishing matched (green), missing (red), and bonus (blue) skills. Based on the computed severity, the application generates a personalized five step action plan with a suggested learning timeline (2-4 weeks low, 4-8 weeks Medium, 10-14 weeks High) and matches each missing skill against a cursed database of 40+ course resources from courser, fast.ai, Kaggle, and HuggingFace, each labelled Free or Paid.

The application is implemented in python using Streamlit >= 1.32, pdfplumber for PDF extraction, python docx, for DOCX parsing and matplotlib for chart generation. The full skill taxonomy (19 technical skills, 16 programming languages, 14 soft skills, 26 tools) is intentionally shared between the ML training pipeline consistency. The app requires only the postings.csv dataset.

VIII. Conclusion & Future work

This paper presented a reproducible benchmark comparing RandomForest, SVM-RBF , and BERT for career readiness and skill gap severity prediction on LinkedIn job posting data. Random Forest achieved the highest macro F1 (0.847) with the second lowest latency (2.3 ms/sample), establishing it as the recommended model for real time advisory deployment. BERT offers competitive accuracy (0.836) but incurs a 21x latency penalty on CPU. The findings were operationalized as a streamlit application that delivers real time, CV generated skill gap analysis to end users without needing ML expertise.

The analysis of feature importance validates that the prevalent predictors of the level of skill gap are the technical skill rating and soft skills.

SMOTE augmented training decreases the High class F1 degradation by 0.61 (no SMOTE) to 0.79.

In future work, real time job matching will be explored by combining live APIs of LinkedIn, Indeed, Glassdoor, and other job boards, the system will combine the uploaded CV of the user with the chosen target role and dynamically surface current, actively hiring vacancies whose skill profiles best align with current competencies of the candidate so that every recommendation reflects current market demand rather than past postings history, and allows candidates to see which jobs they are actually competitive in at Second, ATS optimized the resume generation: the skill gap output will be used, with the LinkedIn generated benchmarks profiles and the rankings of feature importance of the trained random forest model, the system will automatically generate a personalized, role specific resume that will pass the applicant tracking system (ATS) screening. These involve injecting the most weighted keywords found by the classifier, sorting sections in the sequence most preferred by ATS parsers, measuring the skill assertions in the language benchmarked, and calculating a predicted ATS compatibility score, which is directly maximized by increasing the chances of the application of a candidate being shortlisted.

Future directions are also planned to be NER based skill extraction to substitute the use of key-words with matchings, multi language extension of XLM-RoBERT a to non English job markets longitudinal verification of severity predictions with actual employment results and fairness audit to identify and address demographic bias in model classification [11].

IX. Reference

[1] A. Castro-Lopez, F. J. Garcia-Penalvo, and M. A. Conde, "Exploration of employability perceptions with blended multi-criteria decision-making methods," *Education + Training*, vol. 64, no. 2, pp. 259–275, Feb. 2022. [Online]. Available: https://doi.org/10.1108/ET-01-2021-0027.

[2] E. Alonso-Bello, L. E. Santana-Vega, and L. Feliciano-García, "Employability skills of unaccompanied immigrant minors in Canary Islands," *Journal of New Approaches in Educational Research*, vol. 9, no. 1, pp. 15–27, Jan. 2020. [Online]. Available: https://doi.org/10.7821/naer.2020.1.468

[3] F. Levy and R. J. Murnane, The New Division of Labor: How Computers Are Creating the Next Job Market. Princeton, NJ, USA: Princeton University Press, 2004.

[4] A. Desai and A. Kulkarni, "Unleashing the potential of ontology in skill extraction," in Proc. 1st Int. Conf. Artificial Intelligence and Machine Learning Applications for Engineering and Technology (ICAET), IEEE, Mar. 2025, pp. 1–6. doi: 10.1109/ICAET.2025.

[5] C. Romero and S. Ventura, "Educational data mining: A review of the state of the art," *IEEE Transactions on Systems, Man, and Cybernetics, Part C (Applications and Reviews)*, vol. 40, no. 6, pp. 601–618, Nov. 2010. doi: 10.1109/TSMCC.2010.2053532

[6] V.-E. Baciu, J. Stiens, and B. da Silva, "MLino bench: A comprehensive benchmarking tool for evaluating ML models on edge devices," *Journal of Systems Architecture*, vol. 155, p. 103262, Oct. 2024. [Online]. Available: https://doi.org/10.1016/j.sysarc.2024.103262

[7] C. Sun, Y. Yang, J. Men, C. Ju, S. Li, and W. Feng, "Biomedical named entity recognition using BERT in the machine reading comprehension framework," *Journal of Biomedical Informatics*, vol. 118, p. 103799, Jun. 2021. [Online]. Available: https://doi.org/10.1016/j.jbi.2021.103799

[8] G. Dobriţa, S.-V. Oprea, and A. Bâra, "An NLP-driven e-learning platform with LLMs and graph databases for personalised guidance," *Connection Science*, vol. 37, no. 1, p. 2518991, Dec. 2025. [Online]. Available: https://doi.org/10.1080/09540091.2023.2518991

[9] A. Di Battista, S. Grayling, M. Hasselaar, K. Leopold, T. Li, M. Rayner, and S. Zahidi, "Future of Jobs Report 2023," World Economic Forum, Geneva, Switzerland, Tech. Rep., May 2023. [Online]. Available: https://www.weforum.org/reports/the-future-of-jobs-report-2023

[10] N. Sancar and N. Cavus, "Smart skills for smart cities: Developing and validating an AI soft skills scale in the framework of the SDGs," *Sustainability*, vol. 17, no. 16, p. 7281, Aug. 2025. [Online]. Available: https://doi.org/10.3390/su17167281

[11] F. Cai, J. Zhang, and L. Zhang, "The impact of artificial intelligence replacing humans in making human resource management decisions on fairness: A case of resume screening," *Sustainability*, vol. 16, no. 9, p. 3840, Apr. 2024. [Online]. Available: https://doi.org/10.3390/su16093840